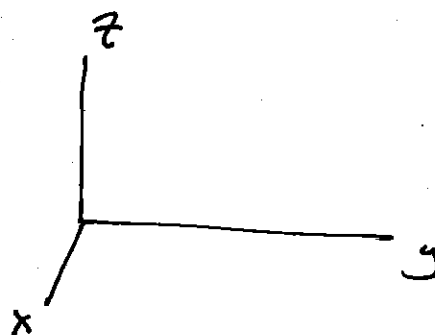
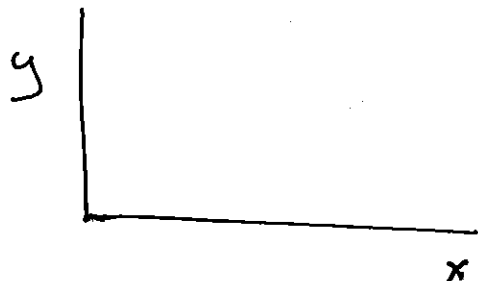


PHYS 1408 CHAPTER 3 VECTORS

1/4

Right hand or Cartesian reference frames



$$(x, y) \rightarrow (r, \theta)$$

$$r = \sqrt{x^2 + y^2}$$

$$\theta = \tan^{-1} \frac{y}{x}$$

$$(r, \theta) \rightarrow (x, y)$$

$$x = r \cos \theta$$

$$y = r \sin \theta$$

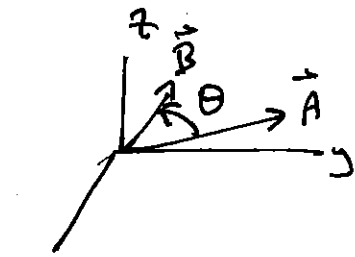
$$\vec{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$$

$$\vec{B} = B_x \hat{i} + B_y \hat{j} + B_z \hat{k}$$

$$\begin{aligned} \vec{C} = 2\vec{A} - \vec{B} &= (2A_x - B_x) \hat{i} + (2A_y - B_y) \hat{j} \\ &\quad + (2A_z - B_z) \hat{k} \end{aligned}$$

Dot product or Scalar product

2/4



$$\vec{A} \cdot \vec{B} = AB \cos \theta \quad \theta \text{ is 3-D angle between } \vec{A} \text{ \& } \vec{B}$$

$$\vec{A} \cdot \vec{B} = A_x B_x + A_y B_y + A_z B_z$$

$$A = |\vec{A}| = \sqrt{A_x^2 + A_y^2 + A_z^2}$$

$$\vec{A} \cdot \vec{B} = AB \cos \theta$$

$$\theta = \cos^{-1} \left(\frac{\vec{A} \cdot \vec{B}}{AB} \right)$$

Cross Product or Vector Product

$$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

$$= \hat{i} \begin{vmatrix} A_y & A_z \\ B_y & B_z \end{vmatrix} - \hat{j} \begin{vmatrix} A_x & A_z \\ B_x & B_z \end{vmatrix} + \hat{k} \begin{vmatrix} A_x & A_y \\ B_x & B_y \end{vmatrix}$$

$$= (A_y B_z - B_y A_z) \hat{i} - (A_x B_z - B_x A_z) \hat{j} + (A_x B_y - B_x A_y) \hat{k}$$

$$\vec{A} = 3\hat{i} + 2\hat{j} - \hat{k}$$

3/4

$$\vec{B} = 6\hat{i} - 6\hat{j} + 2\hat{k}$$

$$\vec{A} \cdot \vec{B} = 3 \cdot 6 + 2(-6) + (-1) \cdot 2$$

$$= 18 - 12 - 2$$

$$= 4$$

$$\theta = \cos^{-1} \frac{\vec{A} \cdot \vec{B}}{AB} = \cos^{-1} \frac{4}{\sqrt{(3^2 + 2^2 + 1^2)} \sqrt{6^2 + 6^2 + 2^2}}$$

$$\theta = \cos^{-1} \frac{4}{\sqrt{14} \sqrt{76}} = \cos^{-1} 0.123 = 82.96^\circ$$

$$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 2 & -1 \\ 6 & -6 & 2 \end{vmatrix} = (4-6)\hat{i} - (6+6)\hat{j} + (-18-12)\hat{k}$$

$$= -2\hat{i} - 12\hat{j} - 30\hat{k}$$

More Accessable 2-D vectors

4/4

$$\vec{A} = 2\hat{i} + 2\hat{j} + 0\hat{k}$$

$$\vec{B} = 0\hat{i} + 3\hat{j} + 2\hat{k}$$

$$\theta = \cos^{-1} \frac{\vec{A} \cdot \vec{B}}{AB} = \cos^{-1} \frac{0 + 6 + 0}{\sqrt{8} \sqrt{13}} = \cos^{-1} 0.588$$

$$\theta = 54.96^\circ$$

$$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 2 & 0 \\ 0 & 3 & 2 \end{vmatrix} = \hat{i} \begin{vmatrix} 2 & 0 \\ 3 & 2 \end{vmatrix} - \hat{j} \begin{vmatrix} 2 & 0 \\ 0 & 2 \end{vmatrix} + \hat{k} \begin{vmatrix} 2 & 2 \\ 0 & 3 \end{vmatrix}$$

$$= 4\hat{i} - 4\hat{j} + 6\hat{k}$$